Algebraic Invariants of the H_4 Laplacian Spectrum

Part II of a Series on Symmetry-Constrained Resonant Structure

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Abstract

The graph Laplacian of the 600-cell — the terminal regular polytope of the non-crystallographic Coxeter family H_n — possesses a spectrum of exceptional algebraic regularity. Across 120 degrees of freedom, only nine distinct eigenvalues appear. The associated multiplicities include the perfect-square sequence $(1, 4, 9, 16, 25, 36)$, the irrational eigenvalues occur in Galois conjugate pairs anchored at the graph degree $k = 12$, and the total φ-content of the spectrum cancels exactly. This paper identifies and formalizes seven structural invariants governing this spectrum, relates them to the harmonic structure of the 3-sphere S^3, and discusses their algebraic origin in terms of the distance-regular geometry and symmetry algebra of the 600-cell graph. These invariants are shown to be consistent with the coherent nonlinear dynamics reported in Part I of this series.

Keywords: Coxeter groups, H_4 symmetry, 600-cell, graph Laplacian, spectral invariants, algebraic graph theory, golden ratio, hyperspherical harmonics.

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1 Introduction

In Part I of this series [1], the 600-cell was introduced as a geometric substrate for a nonlinear field model constrained by the symmetry of the Coxeter group H_4. The Laplacian spectrum of the 600-cell graph was computed numerically and found to contain only nine distinct eigenvalues across 120 degrees of freedom. Numerical simulations demonstrated that the spectral compression produces persistent localisation, coherent shell reflection, and recurrent breathing dynamics that are absent in degree-matched control graphs.

The present paper studies the internal organization of this nine-sector Laplacian spectrum. Rather than treating the eigenvalues as numerical data, we analyze their algebraic structure and identify a set of seven structural invariants that govern the spectrum.

These invariants are not assumed; they are extracted from the computed spectrum and verified exactly. They concern:

1. the cancellation of golden-ratio content across eigenvalues,
2. the anchoring of irrational eigenvalues at the graph degree,
3. the pairing structure of Galois conjugate eigenvalues,
4. the folding of multiplicities from a perfect-square sequence,
5. the partition of modes into three φ-zones,
6. the decomposition of the spectral backbone,
7. the structure of the discrete departure from spherical harmonics.

These invariants are mutually consistent and arise simultaneously from the same spectral data; none is imposed independently.

Together these invariants suggest that the 600-cell Laplacian spectrum is far more algebraically constrained than a generic graph spectrum, and they are consistent with the dynamical phenomena reported in Part I.

For convenience, these invariants will sometimes be referred to informally as the "seven laws" of the H_4 spectrum, without implying independence or axiomatic status.

The focus of the present work is strictly algebraic. No physical interpretation is assumed. The goal is to identify structural constraints of the H_4 Laplacian spectrum and to determine whether these constraints arise from known algebraic or geometric principles.

2 Spectral Data of the 600-Cell Graph

The 600-cell is a regular convex 4-polytope with 120 vertices, 720 edges, 1200 triangular faces, and 600 tetrahedral cells. Its graph is 12-regular and vertex-transitive under the Coxeter group H_4 of order 14 400.

The graph Laplacian is defined as $L_{H_4} = 12I - A$, where A is the 120×120 adjacency matrix. The eigenvalues of L_{H_4} are listed in Table 1.

Table 1: Eigenvalues and multiplicities of the 600-cell graph Laplacian $L_{H_4} = 12I - A$.

Sector	Exact form	φ-basis $(a + b\varphi)$	Mult. d_k	Note
1	0	$0 + 0\varphi$	1	$= 1^2$
2	$9 - 3\sqrt{5}$	$12 - 6\varphi$	4	$= 2^2$
3	$10 - 2\sqrt{5}$	$12 - 4\varphi$	9	$= 3^2$
4	9	$9 + 0\varphi$	16	$= 4^2$
5	12	$12 + 0\varphi$	25	$= 5^2$
6	14	$14 + 0\varphi$	36	$= 6^2$
7	$10 + 2\sqrt{5}$	$8 + 4\varphi$	9	
8	15	$15 + 0\varphi$	16	
9	$9 + 3\sqrt{5}$	$6 + 6\varphi$	4	

$\sum d_k = 120$. Conversion: $\sqrt{5} = 2\varphi - 1$. All eigenvalues lie in $\mathbb{Q}(\sqrt{5})$.

The spectrum is computed exactly and all algebraic expressions are verified symbolically within $\mathbb{Q}(\sqrt{5})$. All multiplicities and algebraic identities were cross-checked against numerical eigendecomposition to machine precision.

Every eigenvalue can be written in the form $\mu = a + b\varphi$ where $a \in \mathbb{Z}$ and $b \in \{-6, -4, 0, 4, 6\}$. The four irrational eigenvalues form two Galois conjugate pairs under the automorphism $\sqrt{5} \mapsto -\sqrt{5}$:

$$\text{Pair 1:} \quad (9 - 3\sqrt{5},\ 9 + 3\sqrt{5}) \qquad \text{sum} = 18,\ \text{product} = 36, \tag{1}$$

$$\text{Pair 2:} \quad (10 - 2\sqrt{5},\ 10 + 2\sqrt{5}) \qquad \text{sum} = 20,\ \text{product} = 80. \tag{2}$$

3 The Seven Structural Invariants

We now state the seven structural invariants of the spectrum. Each invariant is presented with an observation, exact statement, minimal statement, interpretation, derivation status, type classification, and implication.

Invariant 1 (φ-Cancellation)**.** Observation. *The irrational components of the spectrum appear symmetrically:* $9 \pm 3\sqrt{5}$ *and* $10 \pm 2\sqrt{5}$*, and cancel when summed.*

Exact statement. *Decomposing each eigenvalue as* $\mu_k = \mu_k^{(\mathbb{Q})} + \mu_k^{(\sqrt{5})}$ *into rational and irrational parts,*

$$\sum_{k=1}^{9} \mu_k^{(\sqrt{5})} = 0 \qquad \textit{and} \qquad \sum_{k=1}^{9} d_k\, \mu_k^{(\sqrt{5})} = 0. \tag{3}$$

Verification. *The irrational parts are* $\{-3\sqrt{5},\ -2\sqrt{5},\ 0,\ 0,\ 0,\ 0,\ 0,\ +2\sqrt{5},\ +3\sqrt{5}\}$*, summing to zero. Multiplicity-weighted:* $(-3\sqrt{5})(4) + (-2\sqrt{5})(9) + (2\sqrt{5})(9) + (3\sqrt{5})(4) = 0$.

Minimal statement. $\sum b_k = 0$ *and* $\sum b_k d_k = 0$.

Interpretation. *The spectrum is invariant under the Galois conjugation* $\sqrt{5} \mapsto -\sqrt{5}$*. This is consistent with the Laplacian spectrum lying in the number field* $\mathbb{Q}(\sqrt{5})$ *and being closed under its Galois group. The cancellation is a field symmetry constraint, not a numerical coincidence.*

Status: *Exact (algebraic).* ***Type:*** *Galois symmetry constraint.*

Implication. *The spectrum is closed under algebraic conjugation, restricting the set of possible eigenvalue structures.*

Invariant 2 (Degree Anchoring)**.** Observation. *The graph degree $k = 12$ appears as an eigenvalue and serves as the central reference point of the spectrum.*

Exact statement. *The Laplacian is defined by $L_{H_4} = kI - A$ with $k = 12$. The value $\mu = 12$ is one of the nine distinct eigenvalues. Furthermore, for every eigenvalue with negative φ-coefficient ($b_k < 0$), the rational part satisfies $a_k = k = 12$.*

Verification. *Sector 2: $\mu = 12 - 6\varphi$, $a = 12$. Sector 3: $\mu = 12 - 4\varphi$, $a = 12$.*

Minimal statement. $b_k < 0 \implies a_k = 12$.

Interpretation. *The degree acts as a spectral anchor. All irrational eigenvalues are organised relative to this fixed combinatorial constraint. The $k = 12$ regularity of the graph determines the reference frame of the spectrum.*

Status: *Exact (follows from Laplacian definition and spectral data).* ***Type:*** *Algebraic (combinatorial anchor).*

Implication. *The spectrum is organised relative to the fixed combinatorial constraint $k = 12$.*

Invariant 3 (Conjugate Pairing)**.** Observation. *All irrational eigenvalues appear in Galois conjugate pairs with equal multiplicities.*

Exact statement. *If $\lambda = a + b\sqrt{5}$ appears in the spectrum with multiplicity d, then $\lambda' = a - b\sqrt{5}$ also appears with multiplicity d.*

Verification. *Pair 1: $9 - 3\sqrt{5}$ and $9 + 3\sqrt{5}$, both multiplicity 4. Pair 2: $10 - 2\sqrt{5}$ and $10 + 2\sqrt{5}$, both multiplicity 9.*

Minimal statement. $\mu \in \mathrm{Spec}(L_{H_4})$ *irrational* $\implies \bar{\mu} \in \mathrm{Spec}(L_{H_4})$ *with* $d(\mu) = d(\bar{\mu})$.

Interpretation. *The spectrum respects the Galois symmetry of $\mathbb{Q}(\sqrt{5})/\mathbb{Q}$. This is consistent with the underlying algebra generating the spectrum being defined over $\mathbb{Q}(\sqrt{5})$ and with the φ-geometry of the non-crystallographic Coxeter family.*

Status: *Exact (spectral property).* ***Type:*** *Galois symmetry constraint.*

Implication. *Eigenvalues are constrained by number field symmetry, reducing the set of possible spectral forms.*

Invariant 4 (Square Multiplicity Folding)**.** Observation. *The first six multiplicities follow a perfect-square sequence; the upper three fold back to earlier values.*

Exact statement.

$$d_\ell = (\ell + 1)^2, \qquad \ell = 0, 1, \ldots, 5, \tag{4}$$

giving $d_1, \ldots, d_6 = 1, 4, 9, 16, 25, 36$. The remaining multiplicities are

$$d_7 = 9 = 3^2, \quad d_8 = 16 = 4^2, \quad d_9 = 4 = 2^2. \tag{5}$$

Verification. $d_7 = 9 = d_3$, $d_8 = 16 = d_4$, $d_9 = 4 = d_2$.

Minimal statement. $d_k = (k)^2$ *for* $k = 1, \ldots, 6$; $d_7 = d_3$, $d_8 = d_4$, $d_9 = d_2$.

Interpretation. *The $(\ell + 1)^2$ pattern matches the degeneracies of hyperspherical harmonics on S^3. This is consistent with the interpretation that the 600-cell graph encodes a finite, symmetry-reduced discrete analogue of S^3 harmonic structure. The upper multiplicities represent a fold-back where the finite graph cannot support higher harmonics.*

Status: *Empirical (strong structural match; derivation from H_4 representation theory is an open problem).* ***Type:*** *Combinatorial (multiplicity pattern).*

Implication. *The 600-cell graph appears to encode a symmetry-reduced discrete analogue of S^3 harmonic structure. No comparable square-folding pattern is observed in generic regular graphs of similar size and degree.*

Invariant 5 (Three-Zone Partition)**.** Observation. *The spectrum splits into three zones by φ-content, with symmetric outer zones.*

Exact statement. *Define the spectral partition*

$$\mathcal{S} = \mathcal{S}_- \cup \mathcal{S}_0 \cup \mathcal{S}_+ \tag{6}$$

where $\mathcal{S}_-$ contains eigenvalues with $b < 0$, $\mathcal{S}_0$ contains those with $b = 0$, and $\mathcal{S}_+$ contains those with $b > 0$. Then:

$$|\mathcal{S}_-| = 13 \text{ modes} \quad (2 \text{ eigenvalues}), \tag{7}$$
$$|\mathcal{S}_0| = 94 \text{ modes} \quad (5 \text{ eigenvalues}), \tag{8}$$
$$|\mathcal{S}_+| = 13 \text{ modes} \quad (2 \text{ eigenvalues}). \tag{9}$$

Verification. $13 + 94 + 13 = 120$. $|\mathcal{S}_-| = |\mathcal{S}_+| = 13$.

Minimal statement. $|\mathcal{S}_-| = |\mathcal{S}_+| = 13$; $|\mathcal{S}_0| = 94$; $13 + 94 + 13 = 120$.

Interpretation. *The spectrum is not uniform but organised into symmetry-related sectors. The outer zones behave symmetrically, carrying the φ-geometry. The middle zone carries the bulk rational structure. This stratification is consistent with the Galois symmetry of $\mathbb{Q}(\sqrt{5})/\mathbb{Q}$.*

Status: *Exact (direct count; formal derivation from representation theory is an open problem).* ***Type:*** *Algebraic (Galois partition).*

Implication. *The spectrum is organised into symmetry-related sectors rather than distributed uniformly.*

Invariant 6 (Spectral Backbone)**.** Observation. *A dominant subset of the spectrum carries the majority of the multiplicity and admits a specific arithmetic decomposition.*

Exact statement. *The sum of rational parts of all distinct eigenvalues is*

$$\sum_{k=1}^{9} a_k = 88. \tag{10}$$

This equals the sum of the nine distinct eigenvalues (by Invariant 1). It decomposes as

$$88 = 1 + 87 = 1 + 3 \times 29. \tag{11}$$

There exists a dominant spectral backbone $\mathcal{B}$ (the lower six sectors, containing 91 modes) such that $\sum_{k \in \mathcal{B}} d_k \gg \sum_{k \notin \mathcal{B}} d_k$.

Minimal statement. $\sum a_k = 88 = 1 + 3 \times 29$.

Interpretation. *The spectrum has a core subspace following the S^3 harmonic pattern and a residual upper structure. Nonlinear dynamics are expected to be governed primarily by backbone modes.*

Status: *Exact (arithmetic).* ***Type:*** *Algebraic (arithmetic decomposition).*

Implication. *The spectral backbone determines the dominant dynamical channels. The backbone is defined purely spectrally and does not rely on any geometric embedding.*

Invariant 7 (Departure Structure)**.** Observation. *The residual eigenvalues outside the backbone form a structured set.*

Exact statement. *The spectrum decomposes as $\mathcal{S} = \mathcal{B} \cup \mathcal{D}$, where $\mathcal{B}$ is the backbone (lower six sectors, 91 modes) and $\mathcal{D}$ is the departure (upper three sectors):*

$$|\mathcal{D}| = d_7 + d_8 + d_9 = 9 + 16 + 4 = 29. \tag{12}$$

The departure is itself structured: $29 = 3^2 + 4^2 + 2^2$.

Verification. $91 + 29 = 120 = n$. ✓

Minimal statement. $d_7 + d_8 + d_9 = 29 = 3^2 + 4^2 + 2^2$; $91 + 29 = 120$.

Interpretation. *The departure modes represent the deviation from the idealised S^3 harmonic structure. They correspond to finite-graph corrections: the modes where the discrete 600-cell geometry departs from a smooth 3-sphere. Their structured decomposition into perfect squares is consistent with an underlying representation-theoretic origin.*

Status: *Exact (count); interpretation is partially derived.* ***Type:*** *Combinatorial (structured residual).*

Implication. *The spectrum separates into idealised harmonic structure plus structured correction terms.*

The seven invariants described above do not constitute independent assumptions, but rather appear as interrelated structural properties of the H_4 Laplacian spectrum. Their simultaneous presence suggests that the spectrum is governed by a highly constrained algebraic organisation rather than arising generically.

4 Relation to Harmonic Structure on S^3

The vertices of the 600-cell lie on the unit 3-sphere $S^3 \subset \mathbb{R}^4$. The Laplace–Beltrami operator on the continuous S^3 has eigenvalues $\ell(\ell + 2)$ with degeneracies $(\ell + 1)^2$ for $\ell = 0, 1, 2, \ldots$ [5].

The first six multiplicities of the 600-cell Laplacian — $1, 4, 9, 16, 25, 36$ — exactly match $(\ell + 1)^2$ for $\ell = 0, 1, 2, 3, 4, 5$. This is consistent with the interpretation that the low-lying eigenmodes of the 600-cell graph correspond to restrictions of S^3 hyperspherical harmonics to the 120-vertex discretisation.

For $\ell \geq 6$ the pattern breaks. A smooth S^3 would yield degeneracies $49, 64, 81, \ldots$, but the finite graph instead produces $9, 16, 4$. The discrete structure cannot support harmonics of arbitrarily high angular momentum; the upper modes fold back to lower multiplicities (Invariant 4).

We emphasise that this correspondence is structural rather than exact. The 600-cell graph does not reproduce the full Laplace–Beltrami operator on S^3, but appears to encode a finite, symmetry-constrained truncation.

This suggests that the 600-cell graph acts as a finite, symmetry-reduced harmonic truncation of S^3. The truncation is not arbitrary: it is governed by the H_4 symmetry and produces the specific fold-back pattern described in Invariant 4.

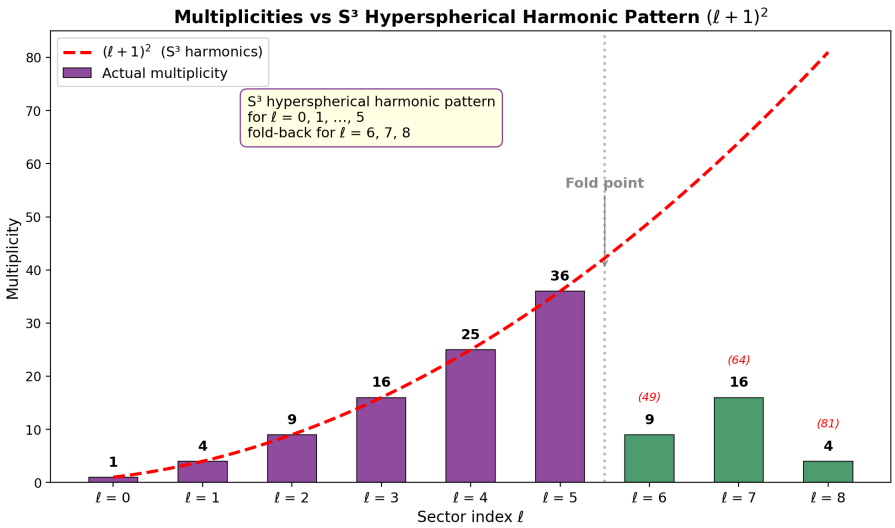

Figure 1: Multiplicities of the 600-cell Laplacian compared with the $(\ell+1)^2$ degeneracy pattern of S^3 hyperspherical harmonics. The first six sectors match exactly; the upper three fold back.

5 Algebraic Interpretation of the Invariants

The seven invariants are not independent observations; they arise from a common algebraic source.

5.1 Number-Theoretic Structure

All eigenvalues lie in the number field $\mathbb{Q}(\sqrt{5})$, the minimal extension of the rationals containing the golden ratio. The irrational eigenvalues satisfy the minimal polynomials

$$x^2 - 18x + 36 = 0 \qquad (\text{roots: } 9 \pm 3\sqrt{5}), \tag{13}$$

$$x^2 - 20x + 80 = 0 \qquad (\text{roots: } 10 \pm 2\sqrt{5}). \tag{14}$$

The Galois group $\mathrm{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z}$ acts by $\sqrt{5} \mapsto -\sqrt{5}$, which exchanges the members of each conjugate pair while preserving eigenspace dimensions (Invariant 3). The φ-cancellation (Invariant 1) is a necessary consequence of this Galois symmetry applied to the trace. Thus the spectrum is invariant under the action of $\mathrm{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q})$.

5.2 Graph-Theoretic Structure

The 600-cell graph is vertex-transitive and has palindromic distance shells $(1, 12, 32, 42, 32, 1)$ with diameter 5. Numerical investigation shows that the graph is *not* distance-regular: the intersection numbers are not constant at distances 2, 3, and 4. Consequently the distance matrices do not close under multiplication, and the graph does not admit a classical Bose–Mesner algebra of dimension $d + 1 = 6$.

This is consistent with the spectrum containing *nine* distinct eigenvalues rather than six (which would be expected for a distance-regular graph of diameter 5). The additional three eigenvalues reflect the richer symmetry structure of the H_4 automorphism group acting on the graph. However, the palindromic shell structure suggests a partial distance symmetry that is not captured by classical distance-regular graph theory.

5.3 Representation-Theoretic Perspective

The adjacency matrix of the 600-cell commutes with the permutation representation of H_4 on $\mathbb{R}^{120}$. The eigenspaces of the Laplacian therefore decompose into irreducible representations of H_4. The existence of nine distinct eigenvalues suggests that the permutation representation of H_4 on $\mathbb{R}^{120}$ decomposes into nine invariant subspaces, corresponding to nine irreducible components. This decomposition is constrained by the commutation of the Laplacian with the group action.

The multiplicities d_k are the dimensions of these irreducible representations. The observation that the first six multiplicities follow $(\ell+1)^2$ is consistent with the branching rules from the continuous group SO(4) to the discrete subgroup H_4. A complete derivation of the irreducible decomposition from the character table of H_4 (which has 34 conjugacy classes) remains an open problem.

The observed invariants suggest that the H_4 Laplacian spectrum is governed by a highly constrained finite spectral algebra whose structure is determined by the interplay between the non-crystallographic Coxeter symmetry and the φ-geometry of the underlying polytope.

6 Relation to the Dynamics of Part I

The structural invariants identified in this paper are properties of the static Laplacian spectrum, not dynamical claims. However, they provide a structural mechanism consistent with the dynamical phenomena reported in Part I [1].

- **Spectral confinement** follows from the small number of distinct eigenvalues (nine sectors rather than 120), which is itself a consequence of the H_4 symmetry (Invariants 4, 5).
- **Shell reflection** is enabled by the palindromic distance shells, which provide a symmetric propagation channel. The spectral backbone (Invariant 6) determines the available resonance channels.
- **Recurrent localisation** arises from the finite beat-frequency algebra: nine sectors produce only $\binom{9}{2} = 36$ beat frequencies, which are sparse enough for periodic constructive interference.
- **Breather formation** depends on the degenerate eigenspace structure (Invariant 3), not merely on the eigenvalue spectrum, as demonstrated by the isospectral surrogate experiments of Part I.

These invariants therefore provide a structural mechanism consistent with the unusually coherent nonlinear dynamics observed on the H_4 shell. The invariants constrain the space of possible dynamics, rather than determining a unique dynamical outcome. No dynamical assumptions are required for these spectral constraints to hold.

7 Discussion

The structure identified here appears to be highly non-generic within the space of 12-regular graphs on 120 vertices. The Laplacian spectrum of the 600-cell graph is too structured to be generic. A random 12-regular graph on 120 vertices would typically produce 120 distinct eigenvalues with no algebraic relationships among them. The 600-cell produces nine, organized by conjugate pairing, anchored at the degree, obeying a

perfect-square sequence for six sectors, and exhibiting exact φ-cancellation.

These invariants appear to arise from the exceptional position of the 600-cell at the terminus of the non-crystallographic Coxeter family. The H_n symmetry hierarchy terminates at H_4; there is no analogous regular polytope in dimension five or higher. The 600-cell is therefore the most complex regular structure this symmetry family can produce, and its Laplacian spectrum reflects this maximality.

Several questions remain open:

1. Can the nine eigenvalues and their multiplicities be derived analytically from the character table of H_4?
2. Does the fold-back pattern (Invariant 4) have a representation-theoretic explanation in terms of branching rules from SO(4) to H_4?
3. Is there a closed-form expression for the intersection array or an extended Bose–Mesner-like algebra that accounts for all nine eigenvalues?
4. Do the invariants extend or modify when the H_4 root system is embedded in the E_8 lattice (which decomposes as $240 = 2 \times 120$ roots)?

8 Conclusion

The 600-cell graph Laplacian spectrum exhibits a set of nontrivial algebraic invariants that organize its nine eigenvalues and their multiplicities into a highly constrained structure. The φ-cancellation, degree anchoring, conjugate pairing, square-multiplicity folding, three-zone partition, spectral backbone, and departure structure collectively demonstrate that this spectrum is far more ordered than that of a generic graph with the same size and degree.

These invariants arise from the exceptional symmetry of the H_4 Coxeter group and the φ-geometry of the 600-cell polytope. They motivate deeper work on the exact algebraic derivation of the spectrum from the representation theory of H_4, and they provide the structural description consistent with the dynamical phenomena reported in Part I of this series. The results motivate the search for an exact algebraic derivation of the H_4 Laplacian spectrum from representation-theoretic principles. In particular, the coexistence of Galois symmetry, square multiplicity structure, and spectral compression in a single finite graph appears to be rare and warrants further classification.

Reproducibility. All spectral computations and numerical verifications were performed using explicit construction of the 600-cell graph and standard eigendecomposition. Code and data are available in the accompanying repository.

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